

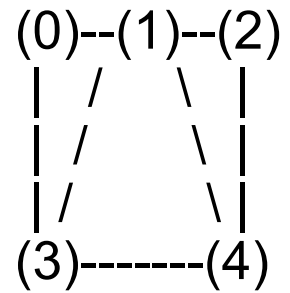


Hamiltonian Cycle

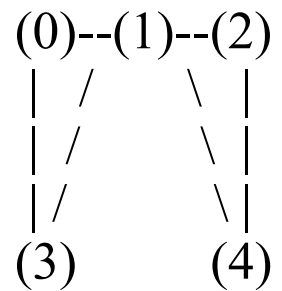
- Hamiltonian cycle is a path in a graph that visits each vertex exactly once and back to starting vertex.
- This program is to determine if a given graph is a hamiltonian cycle or not.
- This program assumes every vertex of the graph to be a part of hamiltonian path.

Example

Hamiltonian Cycle in the following graph is {0, 1, 2, 4, 3, 0}.



And the following graph doesn't contain any Hamiltonian Cycle.



```

1. class HamiltonianCycle{
2.     final int V = 5;
3.     int path[];
4.     boolean isSafe(int v, int graph[][], int path[], int
        pos){
5.         if (graph[path[pos - 1]][v] == 0)
6.             return false;
7.         for (int i = 0; i < pos; i++)
8.             if (path[i] == v)
9.                 return false;
10.        return true;
11.    }
12.    boolean hamCycleUtil(int graph[][], int path[],
        int pos){
13.    if (pos == V){
14.        if (graph[path[pos - 1]][path[0]] == 1)
15.            return true;
16.        else
17.            return false;}

```

```

1.         for (int v = 1; v < V; v++){
2.     if (isSafe(v, graph, path, pos)){
            path[pos] = v;
            if (hamCycleUtil(graph, path, pos + 1) == true)
                return true;
            path[pos] = -1;
            }    }
        return false;    }
        {
            path = new int[V];
            for (int i = 0; i < V; i++)
                path[i] = -1;
            path[0] = 0;
3.        if (hamCycleUtil(graph, path, 1) == false){
4.            System.out.println("\nSolution does not exist");
5.            return 0;
6.        }
7.        printSolution(path);
8.        return 1;}

```

```

1. void printSolution(int path[]){
2.     System.out.println("Solution Exists: Following" +
3.         " is one Hamiltonian Cycle");
4.     for (int i = 0; i < V; i++)
5.         System.out.print(" " + path[i] + " ");
6.     System.out.println(" " + path[0] + " "); }
7. public static void main(String args[]){
8.     HamiltonianCycle hamiltonian =
9.         new HamiltonianCycle();
10.    int graph1[][] = {{0, 1, 0, 1, 0},
11.        {1, 0, 1, 1, 1},
12.        {0, 1, 0, 0, 1},
13.        {1, 1, 0, 0, 1},
14.        {0, 1, 1, 1, 0},
15.    };
16.    hamiltonian.hamCycle(graph1);

```

```

1.    path[0] = 0;
2.    if (hamCycleUtil(graph, path, 1) == false){
3.        System.out.println("\nSolution does not exist");
4.        return 0;
5.    }
6.    printSolution(path);
7.    return 1;
8. }
9. void printSolution(int path[]){
10. System.out.println("Solution Exists: Following" +
11.     " is one Hamiltonian Cycle");
12.     for (int i = 0; i < V; i++)
13.         System.out.print(" " + path[i] + " ");
14. int graph2[][] = {{0, 1, 0, 1, 0},
15.     {1, 0, 1, 1, 1},
16.     {0, 1, 0, 0, 1},
17.     {1, 1, 0, 0, 0},
18.     {0, 1, 1, 0, 0},
19. };
20. hamiltonian.hamCycle(graph2);
21. }

```

/* Let us create the following graph

(0)--(1)--(2)

| / \ |
| / \ |
| / \ |

(3) (4) */

```
int graph2[][] = {{0, 1, 0, 1, 0},
```

```
{1, 0, 1, 1, 1},
```

```
{0, 1, 0, 0, 1},
```

```
{1, 1, 0, 0, 0},
```

```
{0, 1, 1, 0, 0},
```

```
};
```

```
// Print the solution
```

```
hamiltonian.hamCycle(graph2);
```

```
}
```

```
}
```

Output:

Solution Exists: Following is one Hamiltonian Cycle

0 1 2 4 3 0



THANK YOU